SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 1 Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Error Analysis

Given:

- **Training MSE = 0.85**
- **Testing MSE = 5.20**

The large difference between the training and testing Mean Squared Error (MSE) indicates that the neural network performs very well on the training data but poorly on unseen test data. This is a strong sign of **overfitting**.

Possible Reasons

1. **Overfitting**
   - The model has memorized the training data instead of learning general patterns.
   - It fails to generalize to new, unseen data.
2. **Model is Too Complex**
   - The neural network may have too many hidden layers or neurons.
   - A highly complex model can easily fit noise in the training data.
3. **Insufficient Training Data**
   - A small dataset does not represent all possible input patterns.
   - The model learns only the limited examples available.
4. **Presence of Noise or Outliers**
   - Noisy or incorrect data points in the training set may mislead the model.
   - This increases prediction errors on test data.
5. **Lack of Regularization**
   - Techniques such as Dropout or L1/L2 regularization are not used.
   - Without regularization, the model tends to overfit.
6. **Improper Data Splitting**
   - Training and testing datasets may have different distributions.
   - This causes poor performance during testing.

Suitable Corrective Measures

1. **Use Regularization**
   - Apply **L1/L2 regularization** or **Dropout** to reduce overfitting.

2. **Increase Training Data**
    - Collect more data or use **data augmentation** if applicable.
3. **Simplify the Model**
    - Reduce the number of hidden layers or neurons.
    - Use a less complex architecture.
4. **Apply Early Stopping**
    - Stop training when the validation error starts increasing.
    - Prevents the model from memorizing training data.
5. **Perform Cross-Validation**
    - Use techniques such as **k-fold cross-validation** to obtain a more reliable estimate of model performance.
6. **Improve Data Preprocessing**
    - Remove outliers, handle missing values, and normalize or standardize the data before training.
7. **Tune Hyperparameters**
    - Adjust the learning rate, batch size, number of epochs, optimizer, and network architecture for better generalization.

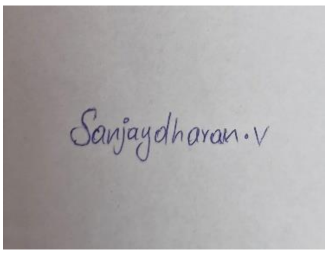
Conclusion

Since the **training MSE (0.85)** is much lower than the **testing MSE (5.20)**, the neural network is **overfitting** the training data. The problem can be reduced by using **regularization techniques, early stopping, increasing the amount of training data, simplifying the model, proper preprocessing, and hyperparameter tuning** to improve generalization on unseen data.

Code of Conduct:

I SANJAY DHARAN V 192425002 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary (5)**	**Level 3 – Proficient (4)**	**Level 2 – Developing (2-3)**	**Level 1 – Beginning (0-1)**
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided